

Information Geometry-based Robust Bayesian Analysis

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Outline

Introduction

Bayesian inference

Information Geometry-based Robust Bayesian Analysis

Numerical examples

Conclusion and perspectives

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Introduction

- ▶ **Bayesian inference** provides a comprehensive probabilistic framework for quantifying epistemic uncertainties and updating them from new information, typically measured data
(Gelman et al., 2013; Kaipio and Somersalo, 2005)
- ▶ It is based on the definition of a **prior distribution**, which summarizes a state of knowledge about some input parameters before making any observation
- ▶ By conditioning on observed data, Bayesian inference enables to derive a **posterior distribution**, reflecting an updated state of knowledge about parameters
- ▶ In this framework, the selection of the prior distribution is of utmost importance !
⇒ Necessity to analyze the impact of the choice of the prior on Bayesian inference results

Introduction

- ▶ **Robust Bayesian analysis** aims at studying the sensitivity of Bayesian inference outcomes with respect to the inputs for that inference (including priors) (Berger, 1990; Ruggeri et al., 2005)
 - ▶ More recently, the emerging field of **robustness analysis** enables the development of methods measuring the impact of the choice of an input distribution, by studying variations of a quantity of interest (QoI) with respect to this choice (Iooss et al., 2020)
 - ▶ In particular, **Perturbed-Law based sensitivity Indices** (PLI) are interesting tools to study variations of a given QoI, for a given perturbation of the input distribution (Lemaître et al., 2015; Sueur et al., 2016)
 - ▶ Recent work provides a rigorous mathematical framework for perturbing input distributions, based on **information geometry** concepts (Gauchy et al., 2022)
- ⇒ **Main goal: definition of information geometry-based PLI dedicated to Bayesian inference**

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General framework of Bayesian inference

- ▶ Let $\mathbf{X}, \mathbf{Y}$ be random variables respectively taking values in $\mathcal{X} \subset \mathbb{R}^d$ and $\mathcal{Y} \subset \mathbb{R}^n$, where:
 - $\mathbf{X}$ represents uncertain input **parameters** of a given model,
 - $\mathbf{Y}$ represents **observables**.
- ▶ Input parameters $\mathbf{X}$ are assumed to admit a density $\pi_{\mathbf{X}}$, known as the **prior** density
- ▶ Given observed data $\mathbf{y} \in \mathcal{Y}$, **Bayes' theorem** enables to derive the following expression for the **posterior** density (i.e., the conditional density of $\mathbf{X}$ knowing $\mathbf{Y} = \mathbf{y}$):

$$\pi_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) = \frac{\mathcal{L}(\mathbf{x})\pi_{\mathbf{X}}(\mathbf{x})}{C}$$

where:

- $\mathcal{L}(\mathbf{x}) = \pi_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x})$ is the **likelihood** (i.e., the conditional density of $\mathbf{Y}$ knowing $\mathbf{X} = \mathbf{x}$, for $\mathbf{x} \in \mathcal{X}$),
- $C > 0$ is the **model evidence**:

$$C = \int_{\mathcal{X}} \mathcal{L}(\mathbf{x})\pi_{\mathbf{X}}(\mathbf{x})d\mathbf{x}.$$

Bayesian inference seen as a reliability problem: the BUS framework

- ▶ Bayesian inference can be reformulated into a *structural reliability* problem (i.e., a failure probability estimation problem), via the **BUS** (*Bayesian Updating with Structural reliability methods*) framework (Straub and Papaioannou, 2015)

- ▶ Let $\Upsilon \sim \mathcal{U}([0, 1])$ independent from $\mathbf{X}$, and let $\mathcal{D}_f \subset \mathcal{X} \times [0, 1]$ be the **BUS failure domain** defined by:

$$\mathcal{D}_f = \{(\mathbf{x}, v) \in \mathcal{X} \times [0, 1] \mid v \leq \beta \mathcal{L}(\mathbf{x})\}$$

where $\beta > 0$ is a constant such that $\beta \mathcal{L}(\mathbf{x}) \leq 1$, for all $\mathbf{x} \in \mathcal{X}$.

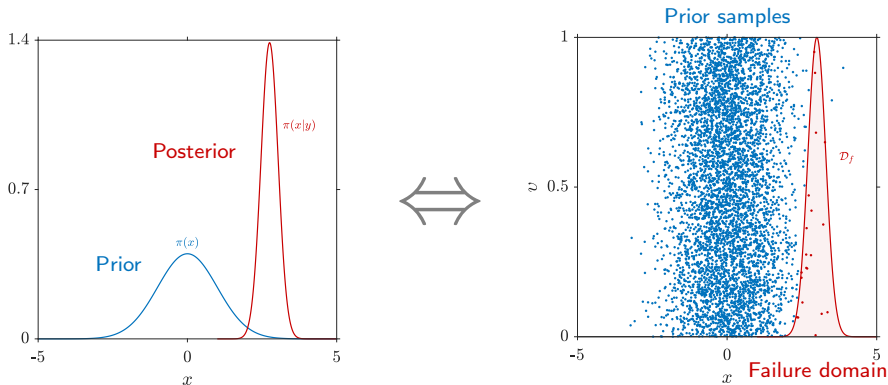
- ▶ One then defines the **BUS failure probability** by :

$$P_f = \mathbb{P}((\mathbf{X}, \Upsilon) \in \mathcal{D}_f) = \int_0^1 \int_{\mathcal{X}} \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}, v) \pi_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} dv$$

Bayesian inference seen as a reliability problem: the BUS framework (bis)

- **Main result:** a realization $(\mathbf{x}, v) \in \mathcal{X} \times [0, 1]$ of $(\mathbf{X}, \Upsilon)$ lies in $\mathcal{D}_f$ if and only if $\mathbf{x}$ follows the posterior distribution:
(Straub and Papaioannou, 2015)

$$(\mathbf{x} \sim \pi_{\mathbf{X}|\mathbf{Y}}(\cdot|\mathbf{y})) \Leftrightarrow ((\mathbf{x}, v) \in \mathcal{D}_f)$$



Bayesian inference seen as a reliability problem: the BUS framework (ter)

- **Main result:** a realization $(\mathbf{x}, v) \in \mathcal{X} \times [0, 1]$ of $(\mathbf{X}, \Upsilon)$ lies in $\mathcal{D}_f$ if and only if $\mathbf{x}$ follows the posterior distribution: (Straub and Papaioannou, 2015)

$$(\mathbf{x} \sim \pi_{\mathbf{X}|\mathbf{Y}}(\cdot|\mathbf{y})) \Leftrightarrow ((\mathbf{x}, v) \in \mathcal{D}_f)$$

- The **BUS failure probability** is linked to model evidence C :

$$P_f = \beta C$$

where β is the BUS scaling constant (chosen such that $\beta \mathcal{L}(\mathbf{x}) \leq 1$, for all $\mathbf{x} \in \mathcal{X}$)

- **Optimal BUS scaling constant:** $\beta^* = 1/\mathcal{L}_{\max}$, where $\mathcal{L}_{\max} = \max_{\mathbf{x} \in \mathcal{X}} \mathcal{L}(\mathbf{x})$ can be estimated with an optimization algorithm

- Recent development of structural reliability methods tailored to BUS framework, in order to sample from the posterior distribution (e.g., rejection sampling, subset simulation)

(Betz et al., 2018; DiazDelaO et al., 2017)

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Fisher distance between densities

- ▶ The prior distribution is assumed to admit independent marginals, and the associated densities $\pi_{X_1}, \dots, \pi_{X_d}$ are assumed to belong to the family of **parametric densities** $\mathcal{S} = \bigcup_{\Theta \subset \mathbb{R}^r, r \geq 1} \{\pi_{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \in \Theta\}$
- ▶ Given a **parametric space** $\Theta \subset \mathbb{R}^r$, with $r \geq 1$, a set of parameters $\boldsymbol{\theta} \in \Theta$ determines a unique density $\pi_{\boldsymbol{\theta}}$ in the manifold $\mathcal{S}_{\Theta} = \{\pi_{\tilde{\boldsymbol{\theta}}} \mid \tilde{\boldsymbol{\theta}} \in \Theta\}$
- ▶ $\mathcal{S}_{\Theta}$ defines a Riemannian manifold when equipped with the **Fisher metric** defined by: (Rao, 1945)

$$\mathcal{I}(\boldsymbol{\theta}) = \mathbb{E}[(\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\mathbf{X})) (\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\mathbf{X}))^{\top}]$$

- ▶ The **Fisher distance** between two densities $\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2} \in \mathcal{S}_{\Theta}$ is then defined by:

$$d_F(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2}) = \min_{\gamma \in \mathcal{P}(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2})} \ell(\gamma)$$

where $\mathcal{P}(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2})$ is the set of paths joining $\pi_{\boldsymbol{\theta}_1}$ to $\pi_{\boldsymbol{\theta}_2}$, and $\ell(\gamma)$ the **length** (in Fisher metric sense) of γ

Fisher perturbations of densities

- ▶ A **geodesic** is a path which minimizes its length (in the Fisher metric sense) (Costa et al., 2015)

- ▶ Given $\pi_{\theta_0} \in \mathcal{S}_\Theta$ and $\theta_0 \in \Theta$, the **Fisher sphere** of center π_{θ_0} and radius $\delta > 0$ is defined by:

$$\partial\mathcal{B}_F(\pi_{\theta_0}, \delta) = \{\pi_\theta \in \mathcal{S}_\Theta \mid d_F(\pi_{\theta_0}, \pi_\theta) = \delta\}$$

i.e., the collection of the endpoints of all geodesics starting from π_{θ_0} with length δ

- ⇒ **Definition of a perturbed density:** given a marginal π_{X_i} and a perturbation level $\delta > 0$, a perturbed density $\pi_{X_i, \delta}$ is defined by a point lying on the Fisher sphere $\partial\mathcal{B}_F(\pi_{X_i}, \delta)$ (Gauchy et al., 2022)

- ▶ Perturbed densities can be determined by computing **geodesics** on the manifold $\mathcal{S}_\Theta$

Computation of geodesics

- ▶ Let $\gamma : [0, 1] \longrightarrow \Theta$ be a path starting from θ_0 and ending at θ_δ , i.e., $\gamma(0) = \theta_0$, $\gamma(1) = \theta_\delta$
- ▶ A sufficient condition for γ to be a geodesic with length $\delta > 0$ is to verify the following system of ordinary differential equations, known as **Hamilton's equations**: (Deriglazov, 2010)

$$\begin{cases} \dot{\gamma} = \nabla_{\zeta} \mathcal{H}(\zeta, \gamma) \\ \dot{\zeta} = -\nabla_{\gamma} \mathcal{H}(\zeta, \gamma) \end{cases}$$

with $\zeta = \mathcal{I}(\gamma)\dot{\gamma}$, and where $\mathcal{H} : \Theta \times \Theta \longrightarrow \mathbb{R}$ denotes the so-called **Hamiltonian operator**:

$$\mathcal{H}(\mathbf{p}, \mathbf{q}) = \frac{1}{2} \mathbf{p}^\top \mathcal{I}(\mathbf{q})^{-1} \mathbf{p}.$$

- ▶ Given initial coordinates $\gamma(0) = \theta_0$, the initial velocity $\zeta(0)$ must satisfy the following equation:

$$\zeta(0)^\top \mathcal{I}(\theta_0)^{-1} \zeta(0) = \delta^2$$

$\Rightarrow$ End coordinates θ_δ are computed by solving Hamilton's equations, from initial conditions $(\gamma(0), \zeta(0))$

Computation of geodesics (bis)

- ▶ Hamilton's equations can be solved numerically by using **time integration** methods
(Amari and Nagaoka, 2007; Leimkuhler and Reich, 2005)
- ▶ Use of the **explicit Euler method**, based on a time discretization $0 = t_0 < t_1 < \dots < t_{N-1} < t_N = 1$ of $[0, 1]$ with a uniform time step $\Delta t = 1/N$:
(Gauchy et al., 2022)

Explicit Euler method for Fisher information geodesics computation

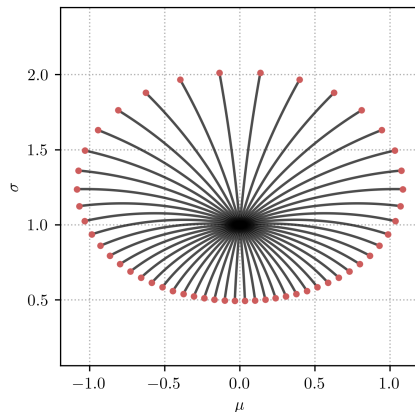
Input: initial density π_{θ_0} , initial velocity ζ_0 , time step $\Delta t = 1/N$

1. Set $\gamma_0 = \theta_0$
2. **For** $i \in \{1, \dots, N\}$:
 - 2.1 Set $\begin{pmatrix} \gamma_i \\ \zeta_i \end{pmatrix} = \begin{pmatrix} \gamma_{i-1} \\ \zeta_{i-1} \end{pmatrix} + \Delta t \begin{pmatrix} \nabla_{\zeta} \mathcal{H}(\zeta_{i-1}, \gamma_{i-1}) \\ -\nabla_{\gamma} \mathcal{H}(\zeta_{i-1}, \gamma_{i-1}) \end{pmatrix}$
3. Set $\theta_{\delta} = \gamma_N$

Output: geodesic endpoint $\pi_{\theta_{\delta}}$

Fisher density perturbations - Example of Gaussian densities

- Example of the manifold of 1D Gaussian densities $\mathcal{S}_\Theta = \{\mathcal{N}(\cdot; \mu, \sigma^2) \mid (\mu, \sigma) \in \Theta\}$, with $\Theta = \mathbb{R} \times \mathbb{R}_{>0}$:



Samples on the Fisher sphere with center $(0, 1)$ and radius $\delta = 1$ (red dots), and associated geodesics (black lines)

Perturbed prior and posterior densities

- ▶ Definition of a perturbed prior joint density, given $i \in \{1, \dots, d\}$ and a perturbation level $\delta > 0$:

$$\pi_{i\delta}(\mathbf{x}) = \left(\prod_{1 \leq j \leq d, j \neq i} \pi_{X_j}(x_j) \right) \pi_{X_{i,\delta}}(x_i),$$

where $\pi_{X_{i,\delta}} \in \partial\mathcal{B}_F(\pi_{X_i}, \delta)$ is a perturbation of the marginal density π_{X_i}

- ▶ The support of the perturbed prior is assumed to be a subset of the unperturbed prior:

$$\text{supp}(\pi_{i\delta}) \subseteq \text{supp}(\pi_{\mathbf{X}})$$

- ▶ The **perturbed posterior** density can then be written as follows:

$$\pi_{i\delta}(\mathbf{x}|\mathbf{y}) = \frac{\mathcal{L}(\mathbf{x})\pi_{i\delta}(\mathbf{x})}{C_{i\delta}}$$

where $C_{i\delta} > 0$ denotes the **perturbed model evidence**:

$$C_{i\delta} = \int_{\mathcal{X}} \mathcal{L}(\mathbf{x})\pi_{i\delta}(\mathbf{x})d\mathbf{x}$$

Perturbed Law-based Indices for Bayesian inference

- Definition of **Bayesian Perturbed Law-based Indices** (BPLI), given a perturbation $\pi_{X_i, \delta} \in \partial \mathcal{B}_F(\pi_{X_i}, \delta)$ with level $\delta > 0$ of the i -th prior marginal density:

$$S(\pi_{X_i, \delta}) = \frac{C_{i\delta} - C}{C}$$

- Definition of minimal and maximal BPLI over the Fisher sphere $\partial \mathcal{B}_F(\pi_{X_i}, \delta)$:

$$S_{i\delta}^- = \min_{\pi \in \partial \mathcal{B}_F(\pi_{X_i}, \delta)} S(\pi) \qquad S_{i\delta}^+ = \max_{\pi \in \partial \mathcal{B}_F(\pi_{X_i}, \delta)} S(\pi)$$

- ⇒ The range $S_{i\delta}^+ - S_{i\delta}^-$ enables to quantify the impact of a lack of knowledge in the prior of X_i on model evidence, for an increasing perturbation level δ

Perturbed Law-based Indices for Bayesian inference (bis)

- By using the BUS framework, BPLI can be reformulated into PLI for failure probabilities:

$$S(\pi_{X_{i,\delta}}) = \frac{P_{i\delta} - P_f}{P_f}$$

where $P_f = \beta C$ denotes the BUS failure probability, and $P_{i\delta}$ the following **perturbed BUS failure probability**:

$$P_{i\delta} = \int_0^1 \int_{\mathcal{X}} \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}, v) \pi_{i\delta}(\mathbf{x}) d\mathbf{x} dv = \beta C_{i\delta}$$

β denoting the BUS likelihood scaling constant

- ⇒ Allows the use of estimation techniques dedicated to PLI for failure probabilities

(Lemaître et al., 2015; Sueur et al., 2016)

Practical computation of BPLI

- ▶ The perturbed failure probability $P_{i\delta}$ can be rewritten as follows:

$$P_{i\delta} = \int_0^1 \int_{\mathcal{X}} \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}, v) \omega_{i\delta}(x_i) \pi_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} dv$$

where $\omega_{i\delta}(x_i) = \frac{\pi_{X_{i,\delta}}(x_i)}{\pi_{X_i}(x_i)}$ denotes the **likelihood ratio**

- ▶ Estimation of $P_{i\delta}$ based on **reverse importance sampling**, from N i.i.d. prior samples $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \sim \pi_{\mathbf{X}}$ and $v^{(1)}, \dots, v^{(N)} \sim \mathcal{U}([0, 1])$:

(Hesterberg, 1996)

$$P_{i\delta} \approx \hat{P}_{i\delta,N} = \frac{1}{N} \sum_{k=1}^N \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}^{(k)}, v^{(k)}) \omega_{i\delta}(x_i^{(k)})$$

- ▶ Estimation of unperturbed failure probability P_f through direct Monte Carlo simulation, from the same prior samples $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \sim \pi_{\mathbf{X}}$:

$$P_f \approx \hat{P}_{f,N} = \frac{1}{N} \sum_{k=1}^N \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}^{(k)}, v^{(k)})$$

Practical computation of BPLI (bis)

- ▶ The BPLI $S(\pi_{X_{i,\delta}})$ is then estimated with the following plug-in estimator:

$$S(\pi_{X_{i,\delta}}) \approx \hat{S}_N(\pi_{X_{i,\delta}}) = \frac{\hat{P}_{i\delta,N} - \hat{P}_{f,N}}{\hat{P}_{f,N}}$$

- ▶ Given one perturbation $\pi_{X_{i,\delta}}$, $S(\pi_{X_{i,\delta}})$ can be estimated from a single set of N prior samples
- ▶ The likelihood ratio $\omega_{i\delta}$ is available in closed form, since the perturbed prior density $\pi_{i\delta}$ is known
- ▶ Nevertheless, it is worth noting that the variance of the reverse importance sampling estimator $\hat{P}_{i\delta,N}$ can be large if the sample size N is not sufficient, especially when δ is large !

Theoretical convergence results for BPLI estimator

Convergence results for BPLI estimator (Rossat, 2026)

Under the two following assumptions:

1. $\text{supp}(\pi_{X_{i,\delta}}) \subseteq \text{supp}(\pi_{X_i})$,
2. $\mathbb{E} [\omega_{i\delta}(X_i)^2] = \int_{\text{supp}(\pi_{X_i})} \frac{\pi_{X_{i,\delta}}(x_i)^2}{\pi_{X_i}(x_i)} dx_i < \infty$,

the BPLI estimator $\hat{S}_N(\pi_{X_{i,\delta}})$ converges almost surely to $S(\pi_{X_{i,\delta}})$, namely:

$$\hat{S}_N(\pi_{X_{i,\delta}}) \xrightarrow[N \rightarrow +\infty]{a.s.} S(\pi_{X_{i,\delta}}).$$

Furthermore, the following convergence in distribution holds:

$$\sqrt{N} \left(\hat{S}_N(\pi_{X_{i,\delta}}) - S(\pi_{X_{i,\delta}}) \right) \xrightarrow[N \rightarrow +\infty]{d} \mathcal{N}(0, \mathbf{d}_s^\top \mathbf{\Gamma}_{i\delta} \mathbf{d}_s),$$

where $\mathbf{d}_s = (-1/P_{i\delta}, P_f/P_{i\delta}^2)^\top$, and:

$$\mathbf{\Gamma}_{i\delta} = \begin{pmatrix} P_f(1 - P_f) & P_{i\delta}(1 - P_f) \\ P_{i\delta}(1 - P_f) & \gamma_{i\delta}^2 \end{pmatrix}, \quad \gamma_{i\delta}^2 = \int_{\mathcal{X} \times [0,1]} \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}, v) \omega_{i\delta}(x_i)^2 \pi_{i\delta}(\mathbf{x}) d\mathbf{x} dv.$$

Theoretical convergence results for BPLI estimator (bis)

- ▶ The previous theoretical convergence results for the BPLI estimator $\hat{S}_N(\pi_{X_{i,\delta}})$ can be used to estimate **asymptotic confidence intervals** (CI)
- ▶ The asymptotic covariance matrix can be consistently estimated with the following estimator:

$$\mathbf{\Gamma}_{i\delta} \approx \hat{\mathbf{\Gamma}}_{i\delta,N} = \begin{pmatrix} \hat{P}_{f,N}(1 - \hat{P}_{f,N}) & \hat{P}_{i\delta,N}(1 - \hat{P}_{f,N}) \\ \hat{P}_{i\delta,N}(1 - \hat{P}_{f,N}) & \hat{\gamma}_{i\delta,N}^2 \end{pmatrix}$$

where:

$$\hat{\gamma}_{i\delta,N}^2 = \frac{1}{N} \sum_{k=1}^N \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}^{(k)}, v^{(k)}) \omega_{i\delta}(x_i^{(k)})^2 \pi_{i\delta}(\mathbf{x}^{(k)})$$

Estimation of extremal BPLI over Fisher spheres

- ▶ Extremal values of BPLI $S_{i\delta}^-$ and $S_{i\delta}^+$ are estimated by discretizing the Fisher sphere $\partial\mathcal{B}_F(\pi_{X_i}, \delta)$ into a finite set of $K \geq 1$ samples $\pi_{X_{i,\delta}}^{(1)}, \dots, \pi_{X_{i,\delta}}^{(K)} \in \partial\mathcal{B}_F(\pi_{X_i}, \delta)$:

$$\hat{S}_{i\delta,N}^- = \min_{1 \leq k \leq K} \hat{S}_N(\pi_{X_{i,\delta}}^{(k)})$$

$$\hat{S}_{i\delta,N}^+ = \max_{1 \leq k \leq K} \hat{S}_N(\pi_{X_{i,\delta}}^{(k)})$$

- ▶ The K perturbed densities $\pi_{X_{i,\delta}}^{(1)}, \dots, \pi_{X_{i,\delta}}^{(K)}$ correspond to K geodesics starting from unperturbed prior parameters θ_0 , with initial velocities $\zeta^{(1)}, \dots, \zeta^{(K)}$
- ▶ These K geodesics are computed through time integration of Hamilton's equation, in an **offline** way (since Fisher spheres only depend on the prior)

Sampling over Fisher spheres

- ▶ Initial velocities $\zeta^{(1)}, \dots, \zeta^{(K)}$ have to be determined to compute the associated geodesics
- ▶ Assuming that the Fisher information metric $\mathcal{I}(\theta_0)$ is definite, one considers the following spectral decomposition:

$$\mathcal{I}(\theta_0)^{-1} = \mathbf{W}\mathbf{\Lambda}\mathbf{W}^\top$$

where $\mathbf{W} \in \mathbb{R}^{r \times r}$ is an orthogonal matrix and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_r)$ with $\lambda_1 \geq \dots \geq \lambda_r > 0$

- ▶ For any $\mathbf{u}_k \in S^{r-1}$ lying on the unit sphere (i.e., $\mathbf{u}_k^\top \mathbf{u}_k = 1$), one sets:

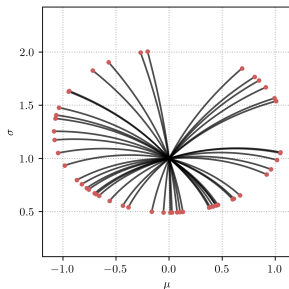
$$\zeta_k = \delta \mathbf{W} \mathbf{\Lambda}^{-\frac{1}{2}} \mathbf{u}_k$$

- ⇒ Then, the initial velocities $\zeta^{(1)}, \dots, \zeta^{(K)}$ are valid, since they are satisfying the initial velocity condition given by:

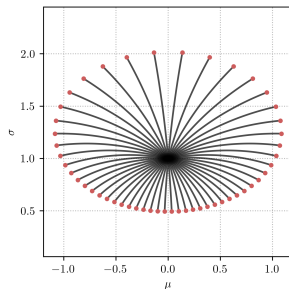
$$\zeta_k^\top \mathcal{I}(\theta_0)^{-1} \zeta_k = \delta^2$$

Sampling over Fisher spheres (bis)

- ▶ Sampling points over the Fisher sphere relies on sampling (normalized) initial velocities over the unit sphere S^{r-1} of $\mathbb{R}^r$
- ▶ In the general case $r \geq 1$, one can use the fact that if $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$, then $\mathbf{u}/\|\mathbf{u}\| \sim \mathcal{U}(S^{r-1})$
 $\Rightarrow$ this possibly lead to a poor coverage of S^{r-1} if K is small, though !
- ▶ In the case $r = 2$, it is possible to directly draw uniformly spaced samples over the circle S^1 :



Geodesics obtained from random sampling on S^1



Geodesics obtained from uniformly spaced points on S^1

Summary of BPLI estimation algorithm

BPLI estimation based on BUS reformulation and reverse importance sampling

Input: prior density $\pi_{\mathbf{X}}$, marginal index $i \in \{1, \dots, d\}$, perturbation level δ , Fisher sphere sample size $K \geq 1$, prior samples $\{\mathbf{x}^{(k)}\}_{1 \leq k \leq N} \sim \pi_{\mathbf{X}}$ and $\{v^{(k)}\}_{1 \leq k \leq N} \sim \mathcal{U}([0, 1])$, indicator evaluations $\{\mathbf{1}_{\mathcal{D}_f}(\mathbf{x}^{(k)}, v^{(k)})\}_{1 \leq k \leq N}$, failure probability estimate $\hat{P}_{f,N}$, BUS scaling constant β

1. Draw K samples $\{\pi_{X_{i,\delta}}^{(k)}\}_{1 \leq k \leq K}$ on the Fisher sphere $\partial \mathcal{B}_F(\pi_{X_i}, \delta)$

2. **For** $k \in \{1, \dots, K\}$:

2.1 Estimate the perturbed BUS failure probability $\hat{P}_{i\delta,N}^{(k)}$ associated to $\pi_{X_{i,\delta}}^{(k)}$:

$$\hat{P}_{i\delta,N}^{(k)} = \frac{1}{N} \sum_{l=1}^N \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}^{(l)}, v^{(l)}) \frac{\pi_{X_{i,\delta}}^{(k)}(\mathbf{x}_i^{(l)})}{\pi_{X_i}(\mathbf{x}_i^{(l)})}$$

2.2 Estimate the PLI $\hat{S}_N(\pi_{X_{i,\delta}}^{(k)}) = (\hat{P}_{i\delta,N}^{(k)} - \hat{P}_{f,N}) / \hat{P}_{f,N}$

3. Set $\hat{S}_{i\delta,N}^- = \min_{1 \leq k \leq K} \hat{S}_N(\pi_{X_{i,\delta}}^{(k)})$ and $\hat{S}_{i\delta,N}^+ = \max_{1 \leq k \leq K} \hat{S}_N(\pi_{X_{i,\delta}}^{(k)})$

Output: BPLI estimates $\hat{S}_{i\delta,N}^-, \hat{S}_{i\delta,N}^+$

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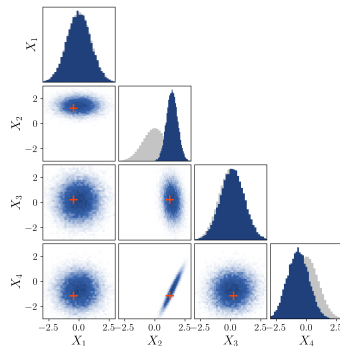
- ▶ Case of a Bayesian inverse problem with input dimension $d = 4$, a standard normal prior $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ and a linear forward model $\mathcal{M}(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$, with $\mathbf{a} = (10^{-6}, 1, 0.1, -0.5)^\top$

- ▶ Likelihood based on an additive Gaussian model with noise $\sigma_0 = 0.1$:

$$Y|\mathbf{X} = \mathbf{x} \sim \mathcal{N}(\mathcal{M}(\mathbf{x}), \sigma_0^2)$$

- ▶ Given data $y \in \mathbb{R}$, the posterior distribution is $\mathcal{N}(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y)$, where:

- $\boldsymbol{\mu}_y = (\boldsymbol{\Sigma}_y \mathbf{a} \sigma_0^{-2}) y$ denotes the posterior mean
- $\boldsymbol{\Sigma}_y = (\mathbf{I}_d + \sigma_0^{-2} \mathbf{a} \mathbf{a}^\top)^{-1}$ the posterior covariance matrix



Posterior distribution (in blue) obtained for the reference prior (in grey) - Samples drawn via Subset Simulation algorithm of (Betz et al., 2018)

Linear Gaussian model - BPLI analytical expression

- ▶ A perturbation of a standard normal prior marginal is fully defined by the following perturbed mean and covariance:

$$\boldsymbol{\mu}_{i,\delta} = \mu_\delta \mathbf{e}_i$$

$$\boldsymbol{\Sigma}_{i,\delta} = \mathbf{I}_d + (\sigma_\delta^2 - 1) \mathbf{e}_i \mathbf{e}_i^\top$$

where $\mu_\delta \in \mathbb{R}$, $\sigma_\delta > 0$ and where $\mathbf{e}_i \in \mathbb{R}^d$ is the i -th vector of the canonical basis of $\mathbb{R}^d$

- ▶ Derivation of a closed form expression for BPLI $S_{i\delta}$:

$$S_{i\delta} = \frac{1}{\sqrt{1 + (\sigma_\delta^2 - 1)r}} \exp \left[\frac{(\sigma_\delta^2 - 1)s^2 + 2\mu_\delta s - \mu_\delta^2 r}{2(1 + (\sigma_\delta^2 - 1)r)} \right],$$

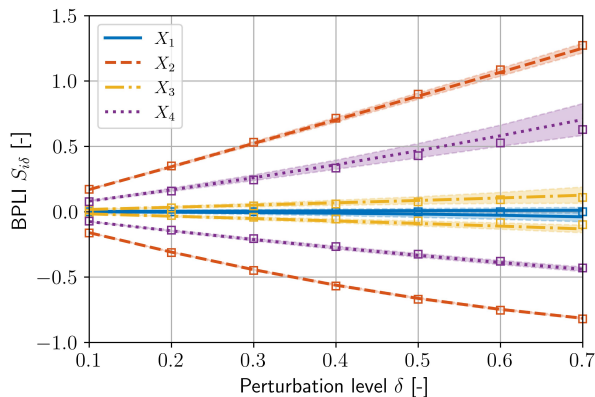
where:

$$r = a_i^2 / (\sigma^2 + \mathbf{a}^\top \mathbf{a})$$

$$s = a_i y / (\sigma^2 + \mathbf{a}^\top \mathbf{a})$$

Linear Gaussian model - BPLI results

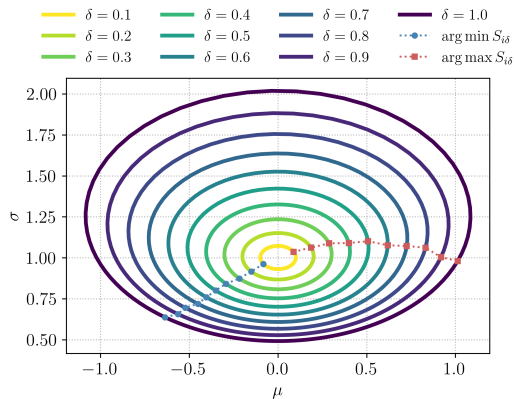
- Estimation of BPLI from $N = 10^5$ prior samples and $K = 100$ Fisher sphere samples



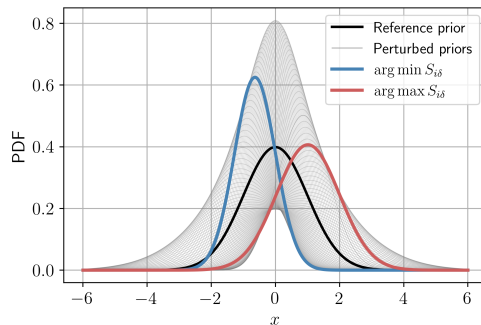
*Minimum and maximum of BPLI over the Fisher sphere of radius δ -
Analytical values are given by squared markers ($\square$)*

Linear Gaussian model - BPLI results (bis)

- Determination of BPLI extrema arguments over Fisher spheres associated to input variable X_2



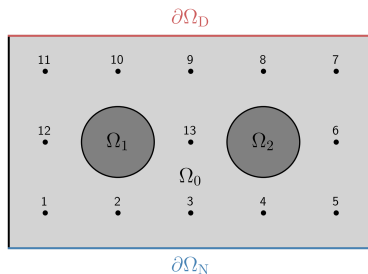
Fisher spheres and BPLI extrema arguments



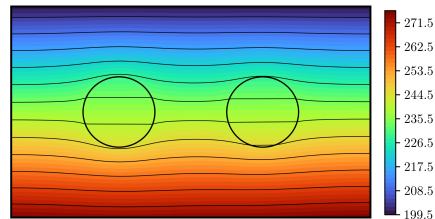
Perturbed prior marginals densities

Inverse heat conduction problem

- ▶ Case of an **inverse heat conduction problem**, involving the inference of thermal conductivities κ_1, κ_2 of two circular inclusions in a rectangular domain, based on temperature measurements (Nagel and Sudret, 2016)
- ▶ Forward model based on 2D heat equation solved with the finite element solver **code_aster**
- ▶ Two cases considered for inference: (a) sensors 1, 2, 10, 11 and 12; (b) all available sensors



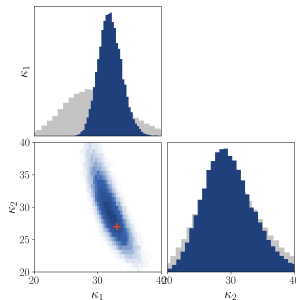
IHCP domain and sensors location



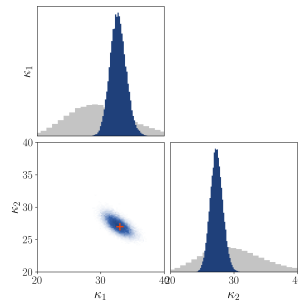
Example of a temperature field

Inverse heat conduction problem

- ▶ Case of an **inverse heat conduction problem**, involving the inference of thermal conductivities κ_1, κ_2 of two circular inclusions in a rectangular domain, based on temperature measurements (Nagel and Sudret, 2016)
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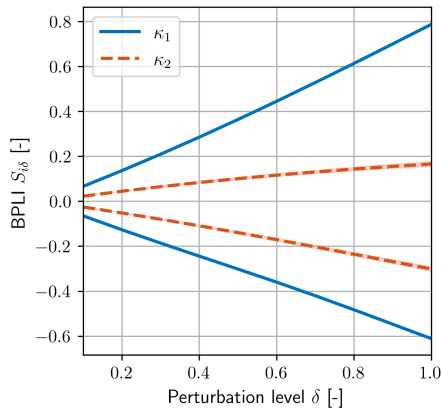
Reference posterior distribution - Case (a)



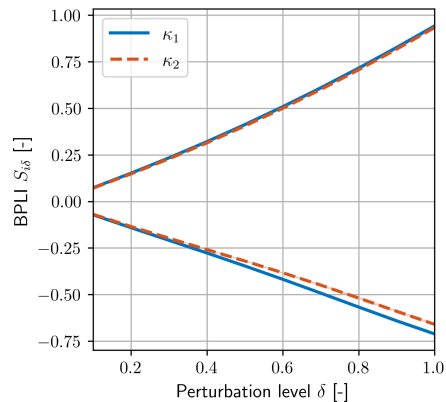
Reference posterior distribution - Case (b)

Inverse heat conduction problem - BPLI results

- Estimation of extremal BPLI over Fisher spheres, from $N = 10^5$ prior samples and $K = 100$ Fisher sphere samples:



Case (a) (sensors around inclusion 1 only)



Case (b) (all sensors)

Nonlinear diffusion inverse problem

- Case of a nonlinear diffusion PDE on a 1D domain $[0, L]$ and a time interval $[0, \tau]$:

$$\partial_t T(\xi, t) = \partial_\xi [\kappa(T(\xi, t)) \partial_\xi T(\xi, t)], \quad \forall \xi \in [0, L], \quad \forall t \in [0, \tau] \quad (\text{Diffusion PDE})$$

$$T(\xi, 0) = 0, \quad \forall \xi \in [0, L] \quad (\text{Initial condition})$$

$$\partial_\xi T(0, t) = q(t) \quad \text{and} \quad \partial_\xi T(L, t) = 0, \quad \forall t \in [0, \tau] \quad (\text{Neumann BC})$$

- Nonlinear expression for diffusivity κ :

$$\kappa(T) = \frac{1}{1 + T^2}$$

⇒ Main goal: infer the time evolution of heat flux q from the temperature history measured at one sensor
(Li and Marzouk, 2014)

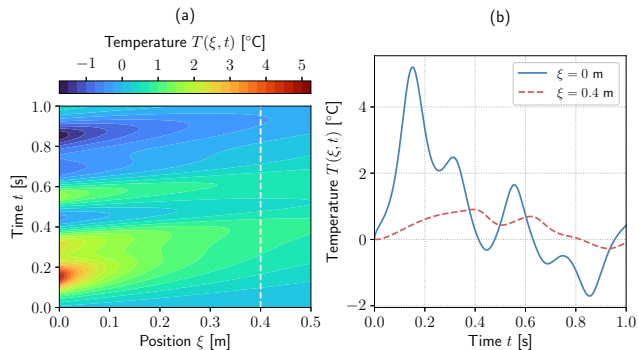
- Heat flux parametrized with truncated Fourier series with N_f modes:

$$q(t) = a_0 + \sum_{j=1}^{N_f} (a_j \cos(2j\pi t/\tau) + b_j \sin(2j\pi t/\tau))$$

⇒ Bayesian inference of Fourier coefficients $\mathbf{x} = (a_0, a_1, b_1, \dots, a_{N_f}, b_{N_f})^\top$ from temperature measurements

Nonlinear diffusion inverse problem - Reference solution

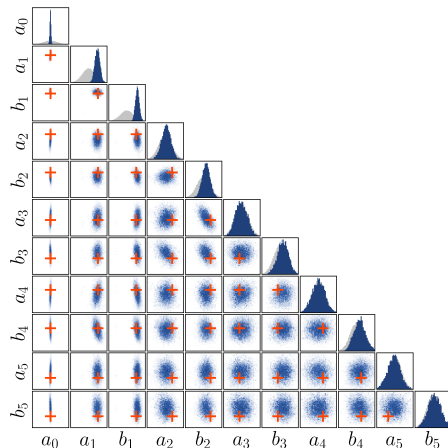
- ▶ Generation of synthetic temperature measurement data, by using $N_f = 5$ Fourier modes ($\Rightarrow$ inverse problem with nominal dimension $d = 11$)
- ▶ Sensor chosen to be quite far away from the boundary with imposed flux $q(t) \Rightarrow$ ill-posed inverse problem



(a) Spatio-temporal evolution of the solution; (b) Detailed view of the temporal evolution of temperature at the left boundary of the domain ($\xi = 0$) and at the temperature sensor location ($\xi = 0.4$).

Nonlinear diffusion inverse problem - Reference solution (bis)

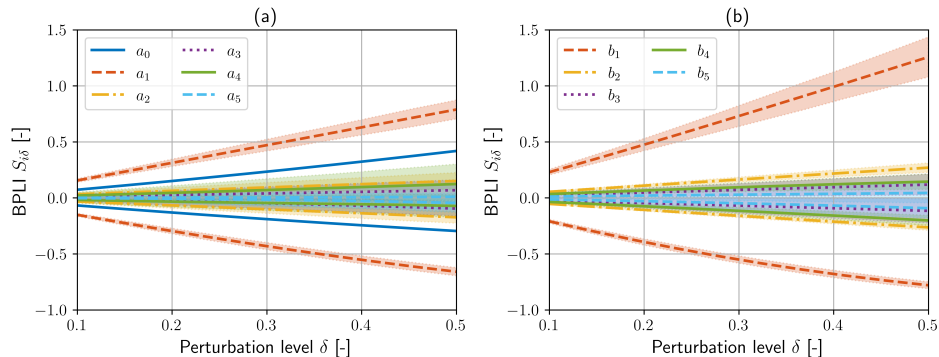
- Estimation of reference posterior distribution with BUS subset simulation algorithm of (Betz et al., 2018):



Comparison of prior and posterior 1D and 2D marginals of Fourier coefficients $(a_0, a_1, b_1, \dots, a_{N_f}, b_{N_f})$. Prior samples are shown in light grey.

Nonlinear diffusion inverse problem - BPLI results

- Estimation of extremal BPLI from $N = 10^6$ prior samples and $K = 100$ Fisher sphere samples:



Minimum and maximum of BPLI over the Fisher sphere, for each Fourier coefficient: (a) a_j , for $j \in \{0, \dots, N_f\}$; (b) b_j , for $j \in \{1, \dots, N_f\}$.

⇒ Results emphasize the low *active dimensionality* of the inverse problem (only first Fourier modes are informed by data)

Outline

Introduction

Bayesian inference

Information Geometry-based Robust Bayesian Analysis

Numerical examples

Conclusion and perspectives

Conclusion

- ▶ Proposition of **Bayesian Perturbed Law-based Indices (BPLI)**, aiming at quantifying the impact of the choice of the prior distribution on results of Bayesian inference (Rossat, 2026)
- ▶ The proposed BPLI can be reformulated into PLI for failure probabilities, for which estimation techniques have been proposed in the recent literature
- ▶ BPLI can help to reduce the dimensionality of Bayesian inverse problems, by focusing on important input parameter
- ▶ BPLI estimation remains tractable to problems involving nonlinear forward models with possibly high-dimensional inputs, while allowing an arbitrary perturbation level for the prior
⇒ contrasts with a significant part of existing Robust Bayesian Analysis approaches (Betrò and Guglielmi, 2000)

Perspectives

- ▶ The likelihood ratio of reverse importance sampling can locally take high values, which can impede the quality of estimated BPLI and their confidence intervals (Gauchy et al., 2022)
- ▶ Similar problems in the case of small BUS failure probabilities (*i.e.* strongly informative data and/or high dimensional input distributions)
⇒ need for more efficient sampling techniques for estimating perturbed BUS failure probabilities (Au and Beck, 2001; Betz et al., 2018)
- ▶ BPLI estimation cost can be prohibitive in the case of costly forward models ⇒ use of **surrogate modeling** techniques (either *prior-based* or *adaptive*) to ensure a feasible computational cost (Marzouk and Najm, 2009; Rossat et al., 2022; Wagner et al., 2021)
- ▶ Possible direct estimation of extremal BPLI on Fisher spheres, by using **optimization methods tailored to Riemannian manifolds** (Ketema, 2026)

Thank you for your attention !

References I

- Amari, S. and Nagaoka, H. (2007). *Methods of Information Geometry*. American Mathematical Society.
- Au, S.-K. and Beck, J. L. (2001). Estimation of small failure probabilities in high dimensions by subset simulation. *Probabilistic Engineering Mechanics*, 16(4):263–277.
- Berger, J. O. (1990). Robust bayesian analysis: sensitivity to the prior. *Journal of Statistical Planning and Inference*, 25(3):303–328.
- Betrò, B. and Guglielmi, A. (2000). *Methods for Global Prior Robustness under Generalized Moment Conditions*, pages 273–293. Springer New York, New York, NY.
- Betz, W., Papaioannou, I., Beck, J. L., and Straub, D. (2018). Bayesian inference with subset simulation: Strategies and improvements. *Computer Methods in Applied Mechanics and Engineering*, 331:72–93.
- Costa, S. I., Santos, S. A., and Strapasson, J. E. (2015). Fisher information distance: A geometrical reading. *Discrete Applied Mathematics*, 197:59–69. Distance Geometry and Applications.
- Deriglazov, A. (2010). *Classical Mechanics*. Springer Berlin Heidelberg.
- DiazDelaO, F., Garbuno-Inigo, A., Au, S., and Yoshida, I. (2017). Bayesian updating and model class selection with subset simulation. *Computer Methods in Applied Mechanics and Engineering*, 317:1102–1121.

References II

- Gauchy, C., Stenger, J., Sueur, R., and Iooss, B. (2022). An information geometry approach to robustness analysis for the uncertainty quantification of computer codes. *Technometrics*, 64(1):80–91.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC.
- Hesterberg, T. (1996). Estimates and confidence intervals for importance sampling sensitivity analysis. *Mathematical and Computer Modelling*, 23(8):79–85.
- Iooss, B., Vergès, V., and Larget, V. (2020). Bepu robustness analysis via perturbed-law based sensitivity indices. In *ANS Best Estimate Plus Uncertainty International Conference (BEPU 2020)*.
- Kaipio, J. and Somersalo, E. (2005). *Statistical and Computational Inverse Problems*. Springer.
- Ketema, B. B. (2026). *Géométrie de l'information pour la robustesse de modèle numérique face à l'incertitude de loi*. PhD thesis, Université de Toulouse.
- Leimkuhler, B. and Reich, S. (2005). *Simulating Hamiltonian Dynamics*. Cambridge Monographs on Applied and Computational Mathematics. Cambridge University Press.
- Lemaître, P., Sergienko, E., Arnaud, A., Bousquet, N., Gamboa, F., and Iooss, B. (2015). Density modification-based reliability sensitivity analysis. *Journal of Statistical Computation and Simulation*, 85(6):1200–1223.

References III

- Li, J. and Marzouk, Y. M. (2014). Adaptive Construction of Surrogates for the Bayesian Solution of Inverse Problems. *SIAM Journal on Scientific Computing*, 36(3):A1163–A1186.
- Marzouk, Y. M. and Najm, H. N. (2009). Dimensionality reduction and polynomial chaos acceleration of bayesian inference in inverse problems. *Journal of Computational Physics*, 228(6):1862–1902.
- Nagel, J. B. and Sudret, B. (2016). Spectral likelihood expansions for bayesian inference. *Journal of Computational Physics*, 309:267–294.
- Rao, C. (1945). Information and the accuracy attainable in the estimation of statistical parameters. *Bull. Calcutta Math. Soc.*, 37.
- Rossat, D. (2026). Information-geometry-based robust bayesian analysis. *International Journal for Uncertainty Quantification*, 16(2):47–73.
- Rossat, D., Baroth, J., Briffaut, M., and Dufour, F. (2022). Bayesian inversion using adaptive polynomial chaos kriging within subset simulation. *Journal of Computational Physics*, 455:110986.
- Ruggeri, F., Ríos Insua, D., and Martín, J. (2005). Robust Bayesian Analysis. In Dey, D. and Rao, C., editors, *Bayesian Thinking*, volume 25 of *Handbook of Statistics*, pages 623–667. Elsevier.

References IV

- Straub, D. and Papaioannou, I. (2015). Bayesian updating with structural reliability methods. *Journal of Engineering Mechanics*, 141(3):04014134.
- Sueur, R., Bousquet, N., Iooss, B., and Bect, J. (2016). Perturbed-law based sensitivity indices for sensitivity analysis in structural reliability. In *Proceedings of the 8th International Conference on Sensitivity Analysis of Model Output (SAMO2016)*, Le Tampon, Réunion Island, France.
- Wagner, P.-R., Marelli, S., and Sudret, B. (2021). Bayesian model inversion using stochastic spectral embedding. *Journal of Computational Physics*, 436:110141.